Buy time before the crash.



A chair-side session twin for dialysis: calibrated forecast, exact fluid arithmetic, and a review pathway.

RESEARCH PROTOTYPE · NOT A PRESCRIPTION

HEMOBP (Taiwan) · 205,482 sessions · 1,070 patients · CC BY 4.0

Fold 2 untouched · no treatment-effect claim · no machine control

THE FIELD REALITY

The doctor is remote.
The technician is at the chair.

We build a remotely operable, IoT-enabled hemodialysis machine. In the units it serves, the person at the chair often has to act before a doctor can be reached.

IDH is one of the most common and risky events of a session. The gap is not another score. It is time and a shared trade-off.

> *"The technician needs to know what to ask, not just that the pressure is falling."*



Labelled hypotension: 10.4% of HEMOBP sessions (nadir <90 mmHg). Spanish cohort: 25.5% clinician-defined symptomatic. Two labels — never combined. 3D concept illustration.

THE GAP

A monitor is not a plan.

CLIMB · 2005

WORSE

Blood-volume monitors plus an algorithm. Outcomes got worse, not better. More data did not equal better care.

OUR EVENT MODEL · 2026

8.2%

Detection at the pre-registered 0.26 false-alert budget — below the current-SBP rule. Ranking is not timing.

THE FIELD

AUROC

Papers report ranking. The chair needs mmHg and a false-alert budget. Prediction without a response workflow fails.

The one deployed system that paired prediction with a nursing response reported IDH 26.1% → 12.9% over four years. The workflow is the product.

Sources: Leung, JAMA 2005; BestShape, Clin Kidney J 2025. Our own event model remains shadow-only.

THE PRODUCT

One console. Three views. One honest boundary.

CASE LAB

Seven chair-side stories, one calibrated chart.

The showpiece.

LIVE SESSION

Enter a real session. Get a protocol state and the fluid arithmetic.

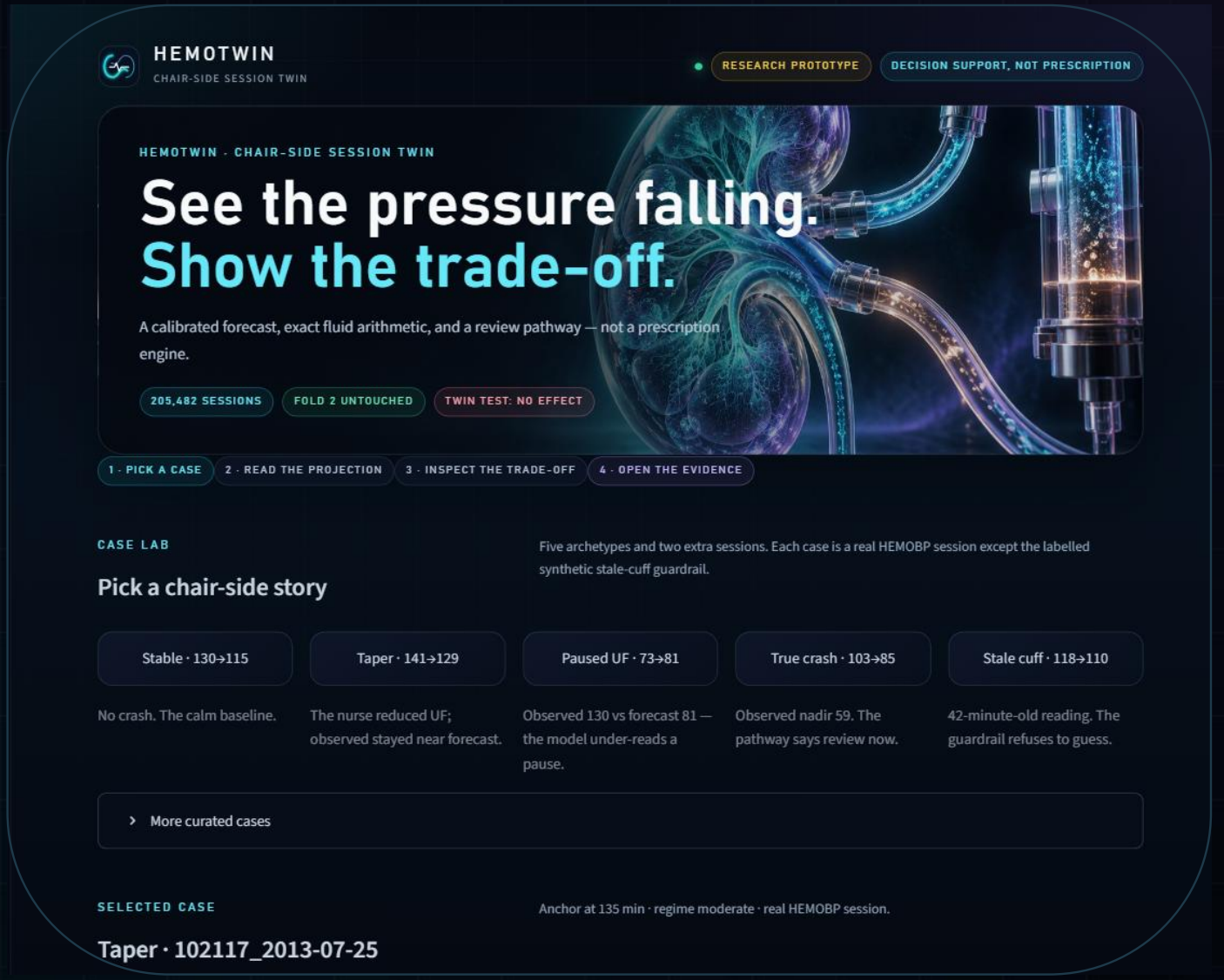
The working console.

EVIDENCE & HANDOFF

Gates, G7, FHIR, audit, pod triage.

The proof.

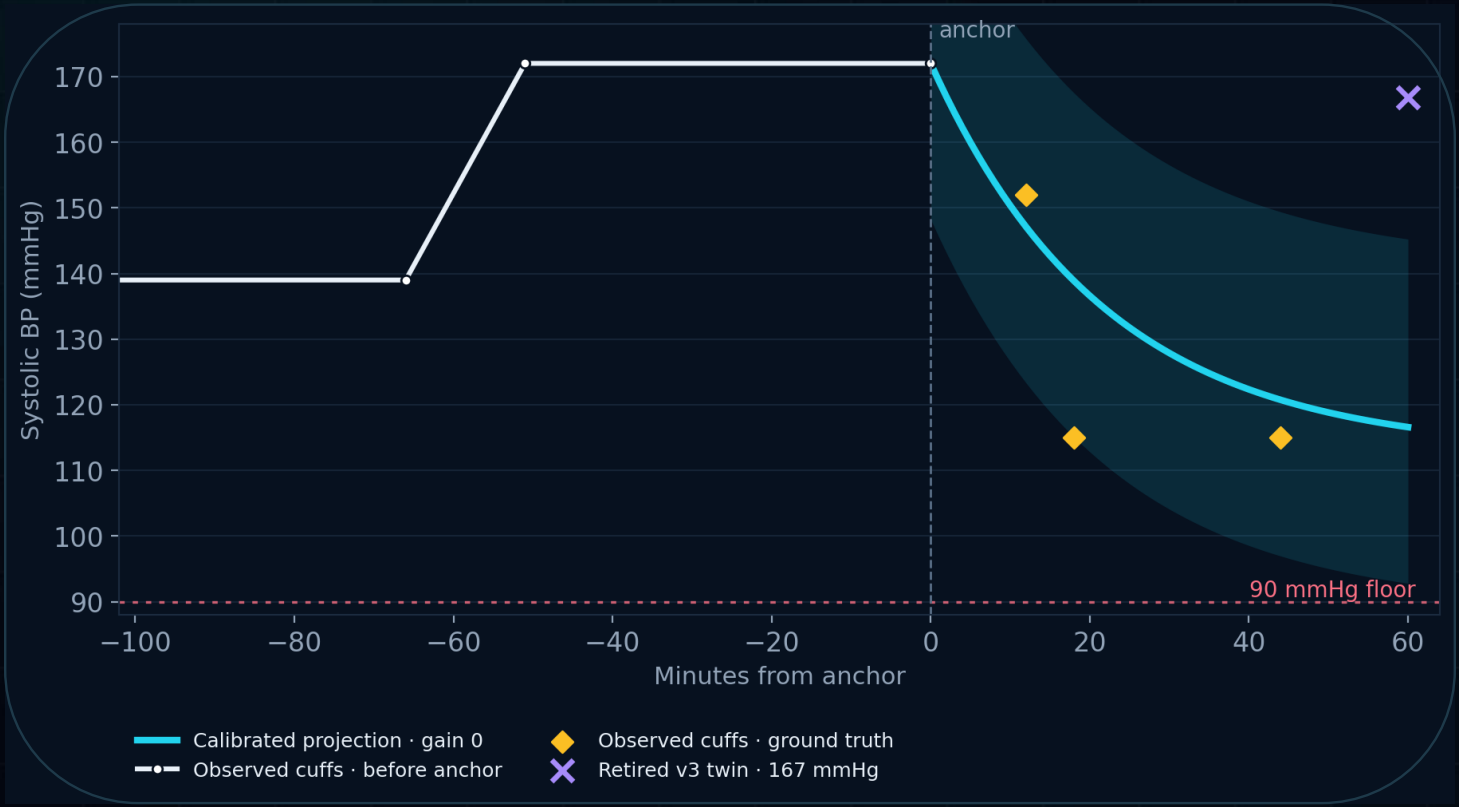
MIRROR · SHOW · BUY TIME



The Case lab, running locally with the bundled evidence. No raw data, no network.

CASE LAB

Seven stories. One honest chart.



g1_taper_showcase · anchor 172 · calibrated 116.6 · observed 115

OBSERVED

115

ground truth · amber diamonds

CALIBRATED

116.6

fair pre-anchor · cyan line

RETIRED V3

167

not used · violet X

Shaded band = 90% nadir interval, widened along the path. The dashed sketch is off by default: the twin experiment failed.

FORECAST LINEAGE

Two forecasts. Do not mix them.

FAIR PRE-ANCHOR TWIN

10.99 fold-2 MAE (mmHg)

- 59 features · no future-UF oracle block.
- RMSE 14.65 · fold 2 untouched.
- This is the calibrated floor used in the console.

USED IN THE CONSOLE

FORECAST-V2 ORACLE DISPLAY

9.60 fold-2 MAE (mmHg)

- 72 features · 13 observed future-UF features.
- Display-only; not a treatment model.
- Not the G2 baseline.

DISPLAY ONLY · NOT THE BASELINE

The smaller oracle number is not a better twin. The baseline check (G2) compares the twin to its own fair baseline; the oracle comparison is a reported failure.

Fold 2 untouched by tuning. G2 plan-written baseline: FAIL.

FLUID ARITHMETIC

The trade-off, in litres.

UF POLICY	FLUID LEFT	NEXT-SESSION UF	CALIBRATED +60
Hold current UF	0.50 L	16.5 mL/kg/h	116.6
Taper to 75%	0.68 L	17.4 mL/kg/h	116.6
Taper to 50%	0.88 L	18.4 mL/kg/h	116.6
Keep-open 0.10 L/h	1.18 L	19.9 mL/kg/h	116.6
30-min pause, then resume	0.88 L	18.4 mL/kg/h	116.6

At calibrated gain 0 the +60 min value is identical across policies. The fluid arithmetic is what changes.

"This is the number a technician can take to a doctor. It is not a prescription."

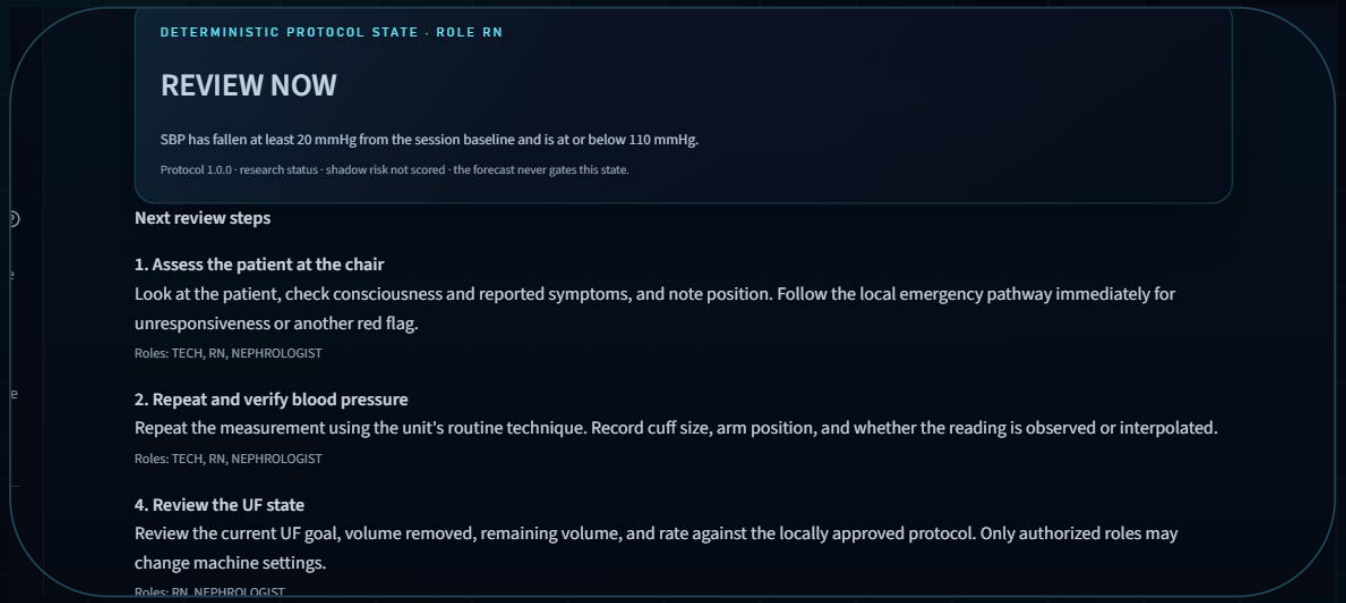
The trade-off is the conversation.

Case g1_taper_showcase: calibrated 116.6 vs observed 115. The fair forecast is close; the arithmetic is the honest decision surface.

Next-session UF reference flag: 13 mL/kg/h. Arithmetic only; not a feasibility, safety, or prescription determination.

DECISION SUPPORT

The next step is a pathway, not a score.



True crash case · observed nadir 59 · protocol state REVIEW NOW

The protocol state is deterministic. The forecast never gates it, and the shadow score has no alert authority. The pathway is the product.

01 **ASSESS THE PATIENT AT THE CHAIR**

Look, check consciousness and reported symptoms, note position. Follow the local emergency pathway for red flags.

02 **REPEAT AND VERIFY BP**

Repeat the measurement with the unit's routine technique. Record cuff size, arm position, observed vs interpolated.

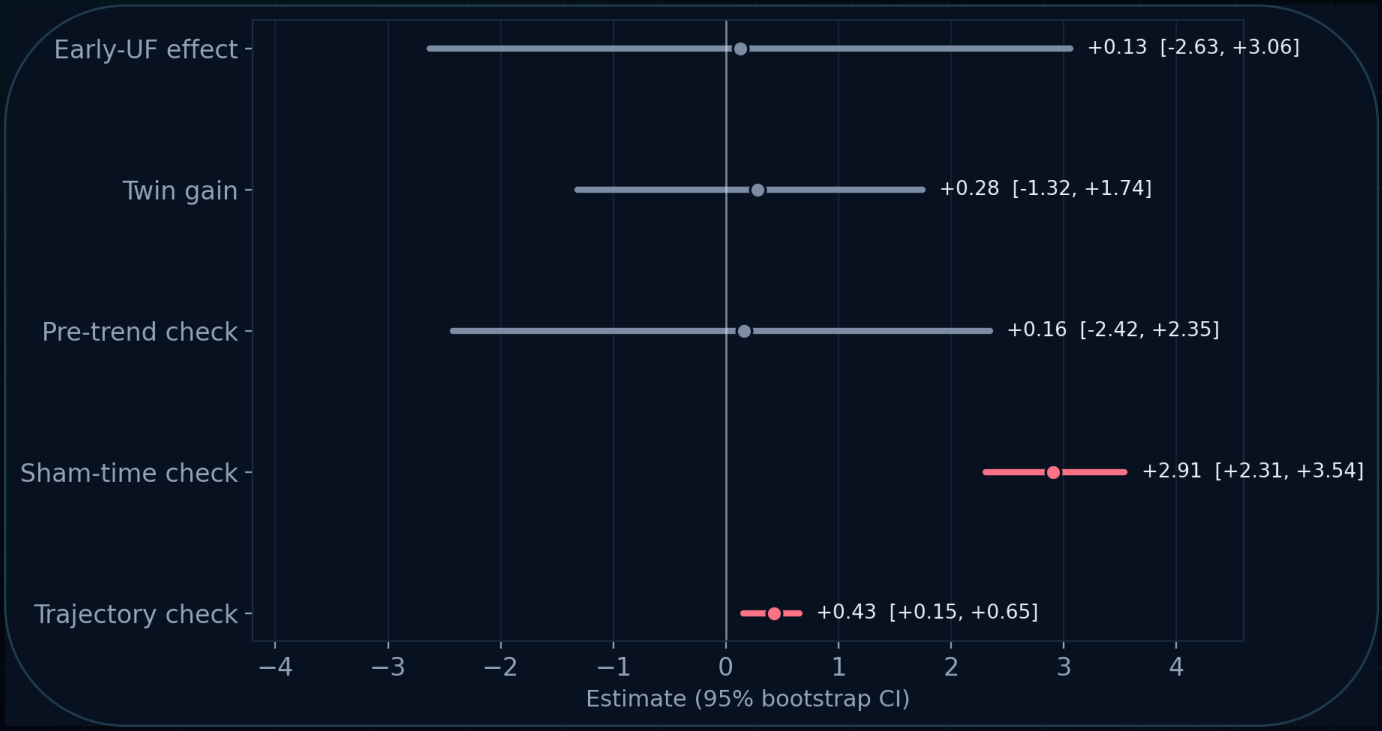
03 **REVIEW THE UF STATE**

Goal, volume removed, remaining volume, rate. Only authorized roles change machine settings.

Measured criteria activate URGENT / REVIEW_NOW / NO_ALERT / INSUFFICIENT. The forecast never gates the state. The shadow score has no alert authority.

THE TWIN EXPERIMENT

We tested it. It failed.



MATCHED EVENTS

581 / 158

events / patients · within-patient

EARLY-UF EFFECT

+0.13

mmHg per 0.25 L/h · CI [−2.63, +3.06]

SHAM-TIME CHECK

+2.91

mmHg · CI [2.31, 3.54] · fails

TRAJECTORY CHECK

+0.43

mmHg worse · CI [0.15, 0.65]

VERDICT: FAIL — early UF reduction did not raise SBP more than matched controls. The effect is indistinguishable from zero; the operational gain stays 0.

Protocol SHA-256 frozen before estimation. 95% bootstrap CIs, patient-clustered.

CLAIM BOUNDARY

What we refuse to claim.

WE CAN CLAIM

- A calibrated pre-anchor nadir forecast: fold-2 MAE 10.99, RMSE 14.65.
- Exact arithmetic for fluid left, carry-forward and next-session intensity.
- A deterministic, versioned review pathway with role-scoped steps.
- Honest negative results: G3, G7 and the oracle G2 comparison.
- A local FHIR export and a hash-chained demo audit trail.

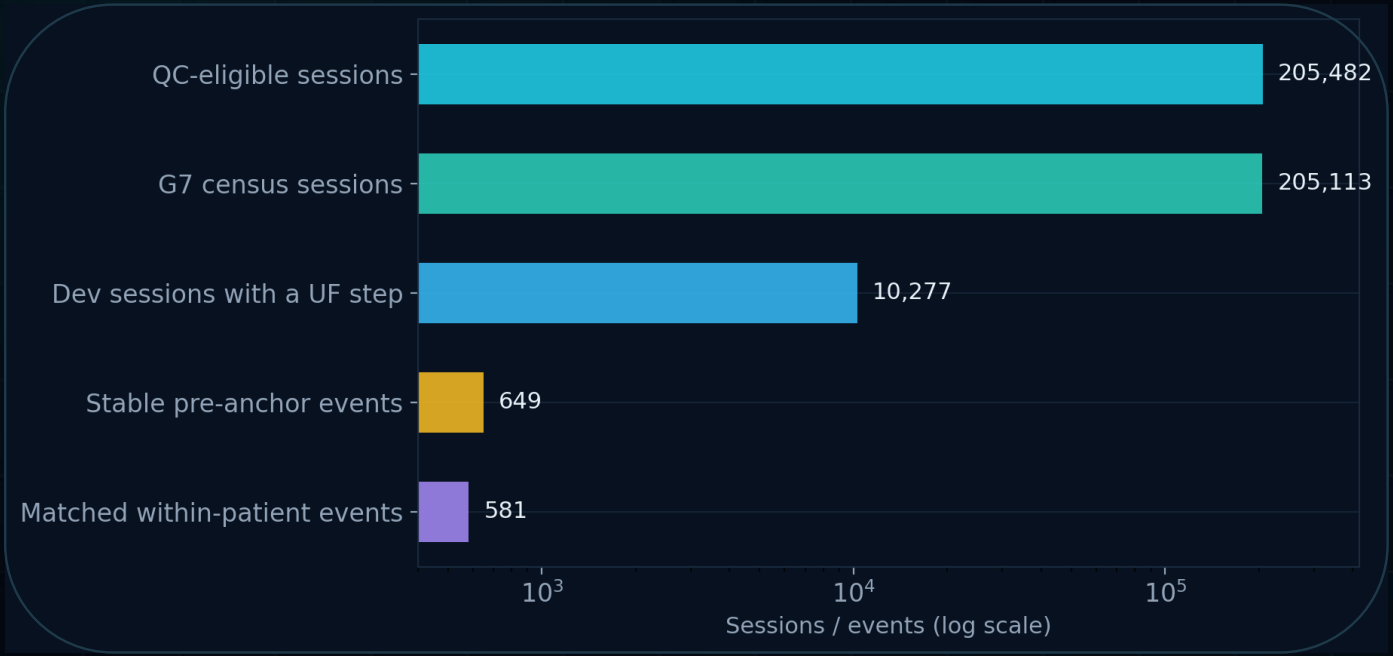
WE REFUSE TO CLAIM

- A validated what-if/treatment effect from the twin.
- That any UF policy will raise SBP, prevent IDH, or is minimum effective.
- Alert utility, clinical benefit, or a safety layer.
- Prospective, external, symptomatic-IDH, or Indian-unit validation.
- A prescription, a default dose, or autonomous machine action.

The boundary is part of the product. It is why the rest is trustworthy.

THE EVIDENCE

205,482 sessions. 581 matched events.



205,482 QC-eligible → 205,113 G7 census → 581 matched events. 369 sessions lack the required pre-anchor record.

SESSIONS
205,482
QC-eligible

PATIENTS
1,070
de-identified

TELEMETRY
4.19M
rows

LABELLED IDH
10.4%
nadir <90 mmHg

FOLD 2 UNTOUCHED

G7 census 205,113 — 369 sessions lack the required pre-anchor record. 25.5% Spanish clinician-defined symptomatic hypotension: a different label, never combined. External transport AUROC 0.474 failed. HEMOBP CC BY 4.0.

ENGINEERING

Built to be inspected.

217

package tests · 0 failures

28 heavy-artifact tests explicitly skip-gated. Full lab suite: 241.

14/14

claim-boundary checks

Frozen G1–G7, banned claims, cohort/demo integrity, deck/README links.

FHIR R4

local export

Bilingual handoff card. No network call at runtime.

SHA-256

audit chain

Demo-only local chain. Not immutable, not authenticated, not clinical.

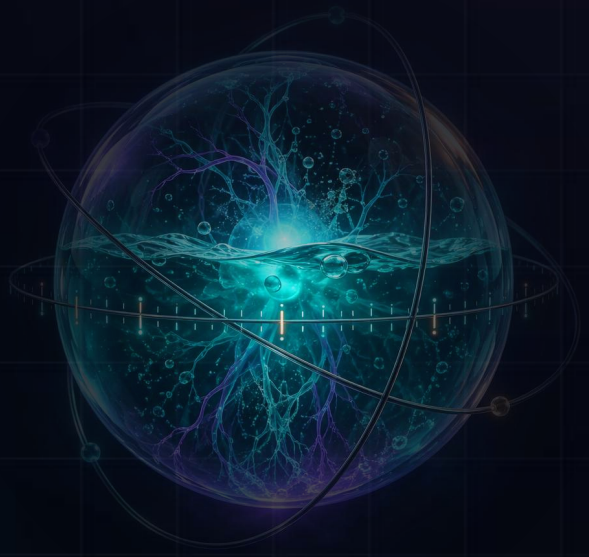
DATA → FEATURES → FORECAST → PROTOCOL → HANDOFF

One app, one README, one deck. The retired v3 console is not shipped; the current console runs from a clean V2 folder with the bundled models and evidence.

THE ASK

Give us a chair, not a stage.

We are not asking for a trial on day one. Run the console beside the team, log what it would have shown, and measure whether it buys time to react.



01

SILENT MODE

2–4 weeks beside the team. No machine control, no alert authority. Log every display.

02

LOCAL RECALIBRATION

Your machine, your population, your protocol. Re-fit the forecast before any claim.

03

PROSPECTIVE EVALUATION

Time-to-review, documentation, and workflow. Pre-register the endpoint before the pilot.

Built by an engineer working on a remotely operable, IoT-enabled hemodialysis machine.

THE NEXT EXPERIMENT IS A WORKFLOW

The twin knows what it cannot do.



A chair-side decision-support layer for dialysis. Calibrated forecast, exact fluid arithmetic, and a review pathway.

Research prototype. Not a medical device. No treatment recommendation.

HemoTwin · one console · three views · one honest boundary

APPENDIX A

Gate definitions and results.

G1	MAE_v4 ≤ 0.80×MAE_v3 and |bias| ≤ 3	PASS	11.59 vs 15.55 · n=374
G2	vs fair base / vs oracle	PASS vs base · FAIL vs oracle	10.99 vs 9.60
G3	actual < hold · CI upper < 0	FAIL	397 rows · Δ 0 [0, 0]
G4	monotone + continuous at unit gain	PASS	0 violations / 3.09M checks
G5	90% band coverage 88–92%	PASS	89.2% HOLD · 90.1% TAPER
G6	cold start + future invariance	PASS	284 · 200/200
G7	does early UF reduction raise SBP?	FAIL	τ +0.13 [−2.63, +3.06]

A pass is an engineering check against a pre-registered criterion — not clinical validation. G2 is a tie at gain 0, not evidence of a treatment effect. G7 asks whether early UF reduction raised SBP; it did not.

APPENDIX B

Q&A ammunition.

IF THEY ASK ABOUT THE MODEL

3 patient folds; fold 2 untouched; 59 pre-anchor features; fold-2 MAE 10.99; 90% band coverage 89–90%; manual sessions are OOD-labelled.

IF THEY ASK ABOUT THE TWIN

The twin test (G7) found no effect: the CI includes zero; sham +2.91; P6a +0.43; operational gain 0.

IF THEY ASK ABOUT THE ALERT

AUROC 0.880 ranking; 8.2% detection at 0.26 false alerts/session; below the current-SBP rule; shadow only, no alert authority.

IF THEY ASK ABOUT DEPLOYMENT

No auth, no machine integration, no prospective or external validation; the local JSONL audit is demo-only, not clinical.

Every number in this deck is traceable to evidence/ and docs/EVIDENCE_STATUS.md.